

Cluster-forming t > 3.98 (p < 1e-4) · cluster-extent FWE p < .05

model7 · n = 58 · surviving extent, % of mask: RPE +37.5 / -0.1 uRPE +1.0 / -8.6 Surprise +0.4 / -0.0

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a

			Covered by map						Effect in ROI			Are they different?		
			RPE		uRPE		Surprise		RPE	uRPE	Surprise	uRPE vs Surprise	RPE vs uRPE	RPE vs Surprise
			+	-	+	-	+	-						
RPE+ & uRPE- * — 6 clusters														
1	Caudate / putamen R	found by RPE +	■	·	·	□	·	·	▲	▼	△	▼	▲	▲
2	Caudate / putamen R (2)	found by uRPE -	■	·	·	□	·	·	▲	▼	△	▼	▲	▲
3	Cingulate gyrus L	found by uRPE -	■	·	·	□	·	·	▲	▼	△	▼	▲	▲
4	Visual cortex (mid/lat occipital) L	found by uRPE -	■	·	·	□	·	·	▲	▼	△	▼	▲	▲
5	Middle temporal gyrus R	found by uRPE -	■	·	·	□	·	·	▲	▼	△	▼	▲	▲
6	Postcentral gyrus R	found by uRPE -	■	·	·	□	·	·	▲	▼	△	▼	▲	▲
RPE+ & uRPE- & Surprise+ * — 4 clusters														
7	DLPFC L	found by uRPE -	■	·	·	□	·	·	▲	▼	△	▼	▲	▲
8	Superior temporal gyrus L	found by Surprise +	■	·	·	□	■	·	▲	▼	△	▼	▲	▲
9	DLPFC L (2)	found by Surprise +	■	·	·	□	■	·	▲	▼	△	▼	▲	▲
10	Angular gyrus R	found by Surprise +	■	·	·	□	■	·	▲	▼	△	▼	▲	▲
RPE- & uRPE+ * — 1 cluster														
11	Insula R (2)	found by RPE -	·	□	■	·	·	·	▼	▲	△	▲	▼	▼
only uRPE+ — 5 clusters														
12	dmPFC R	found by uRPE +	·	·	■	·	·	·	▼	▲	△	▲	▼	○
13	Superior frontal sulcus R	found by uRPE +	·	·	■	·	·	·	▼	▲	△	▲	▼	—
14	Frontal middle gyrus R	found by uRPE +	·	·	■	·	·	·	△	▲	▼	▲	—	—
15	Insula R	found by uRPE +	·	■	■	·	·	·	▼	▲	▼	▲	▼	—
16	Insula L	found by uRPE +	·	·	■	·	·	·	▼	▲	▼	▲	—	○
only uRPE- — 2 clusters														
17	Inferior parietal lobule R	found by uRPE -	·	·	·	□	·	·	△	▼	△	▼	▲	○
18	Inferior parietal lobule L	found by uRPE -	■	·	·	□	·	·	△	▼	▼	▼	▲	—
RPE+ & Surprise+ — 1 cluster														
19	Precuneus L	found by Surprise +	■	·	·	■	■	·	▲	▼	△	▼	▲	—
RPE+ & uRPE+ — 1 cluster														
20	Inferior parietal lobule R (2)	found by uRPE +	■	·	■	·	·	·	▲	▲	▼	▲	○	▲
only RPE+ — 1 cluster														
21	Inferior frontal gyrus L	found by RPE +	■	·	·	·	·	·	▲	△	△	○	—	▲

Profile groups — 8 of them across 21 ROIs

RPE+ & uRPE- * (6) RPE+ & uRPE- & Surprise+ * (4) RPE- & uRPE+ * (1) only uRPE+ (5) only uRPE- (2)

RPE+ & Surprise+ (1) RPE+ & uRPE+ (1) only RPE+ (1)

* the same tissue responds positively to one signal and negatively to another — these groups are listed first.

Membership is map coverage: a map counts if it covers at least a tenth of the ROI. It is not the difference tests — a signal can be in the tag and still be “undetermined” against another.

Rows	Covered by map	Effect in ROI	Are they different?
One cluster found by one of the six contrasts. Rows are boxed by profile — the set of maps covering at least a tenth of them — and the box is drawn in that profile's colour. The number keys the row to its peak on the map pages.	Share of the ROI's voxels inside that map's surviving clusters.	Group one-sample t of that signal's contrast value, averaged over the ROI.	One signal against another inside the ROI, paired across subjects.
	100% of the ROI 50% of the ROI 10% of the ROI Open square = negative tail · Dot = below 1%, no overlap	▲ Positive, and significant ▼ Negative, and significant △ Open = not significant Bigger marker = a larger effect. Holm-corrected over all ROI × signal tests.	▲ Different; points to the larger ○ Equivalent ($BF_{01} > 3$) — Undetermined TOST bound in the table is $d_z = 0.368$, the smallest effect this design can find with 80% power.

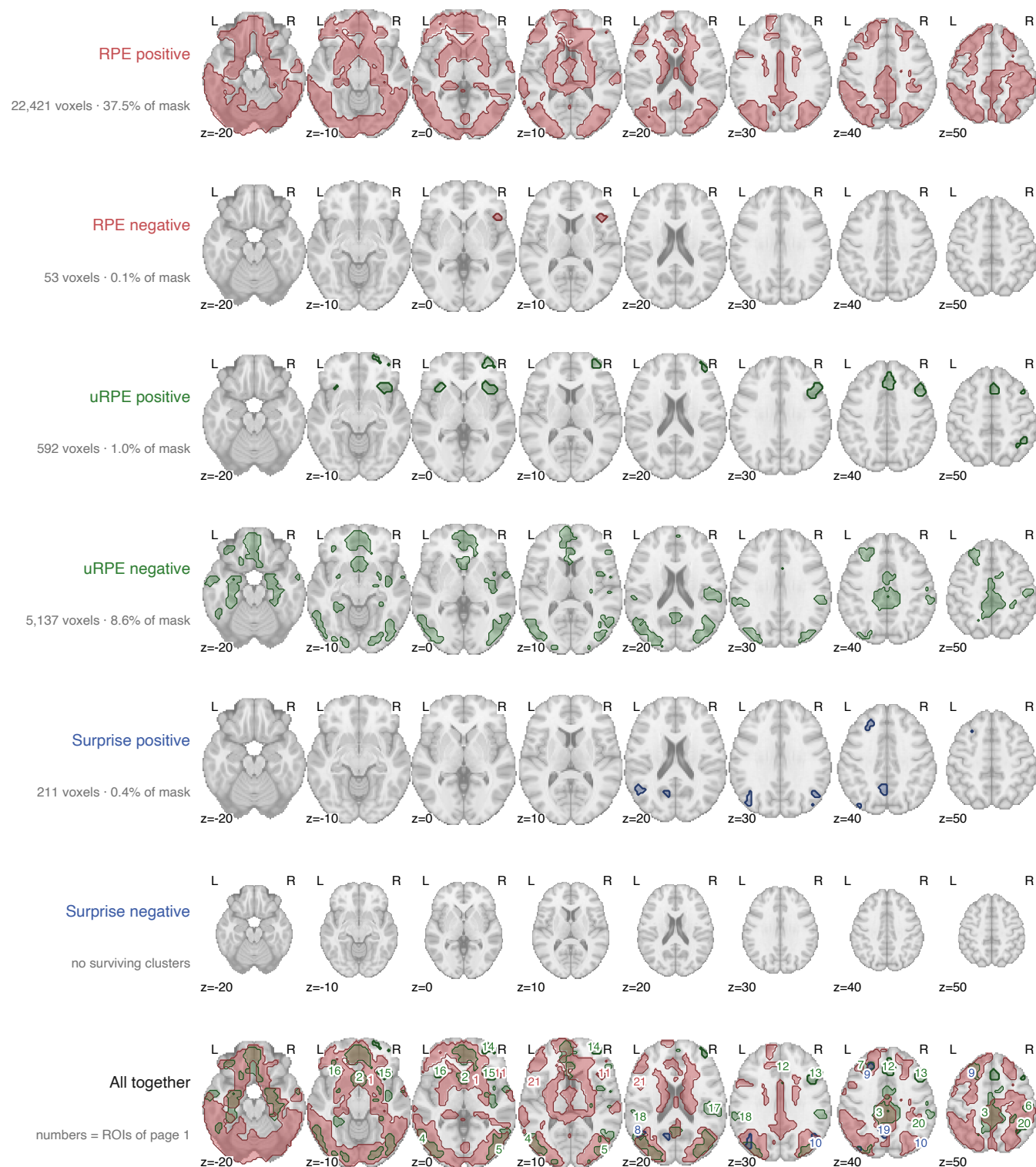
Cluster-forming $t > 3.98$ ($p < 1e-4$) · cluster-extent FWE $p < .05$

model7 · $n = 58$ · surviving extent, % of mask: RPE +37.5 / -0.1 uRPE +1.0 / -8.6 Surprise +0.4 / -0.0

model7 — one GLM carrying all three signals: choice (surprise, value), feedback (uRPE first, then signed RPE). Because SPM orthogonalises serially within an event, uRPE keeps all variance it shares with signed RPE, and the RPE map is the residual. Apples-to-apples across signals.

b Axial sections · $z = -20$ to $+50$ mm

one row per map, then all of them together



Same axial cuts in every row, so extent can be compared straight down a column.

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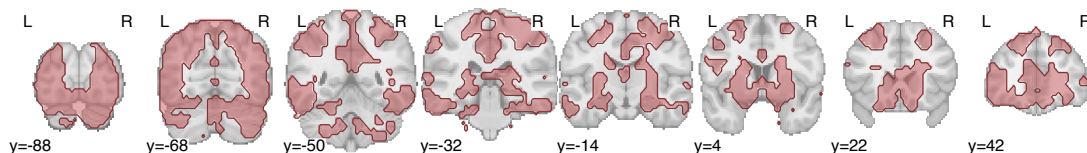
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c Coronal sections · $y = -88$ to $+42$ mm

one row per map, then all of them together

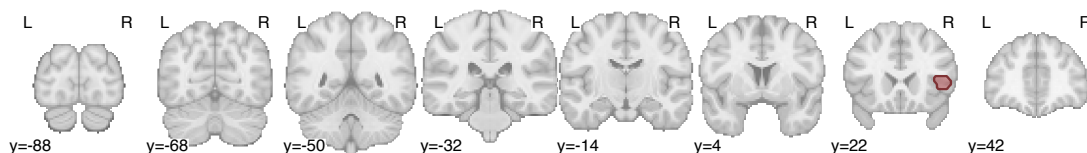
RPE positive

22,421 voxels · 37.5% of mask



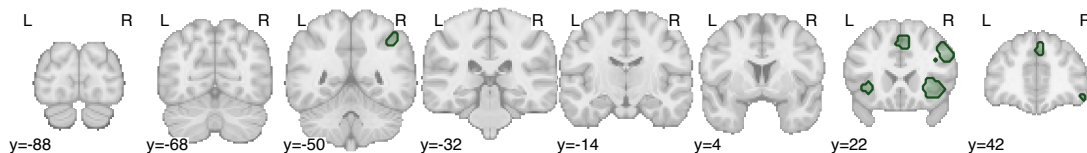
RPE negative

53 voxels · 0.1% of mask



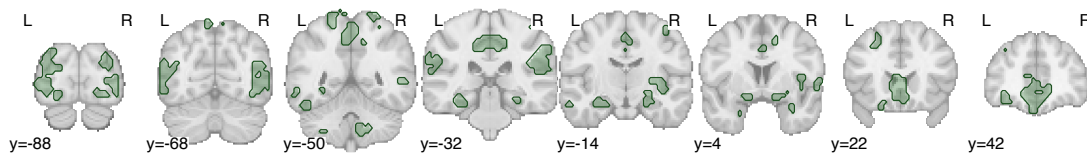
uRPE positive

592 voxels · 1.0% of mask



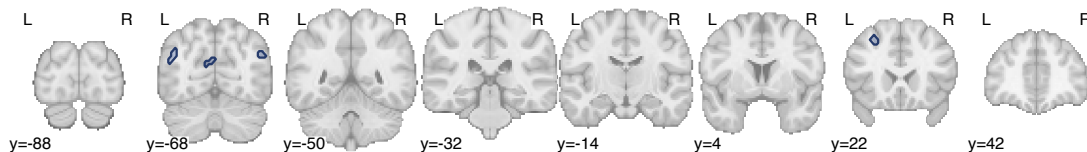
uRPE negative

5,137 voxels · 8.6% of mask



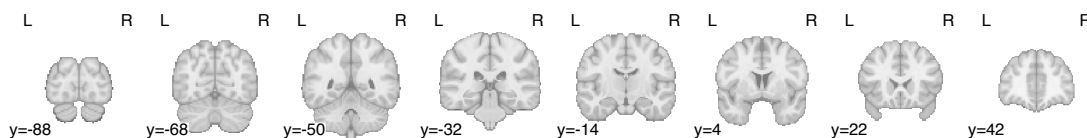
Surprise positive

211 voxels · 0.4% of mask



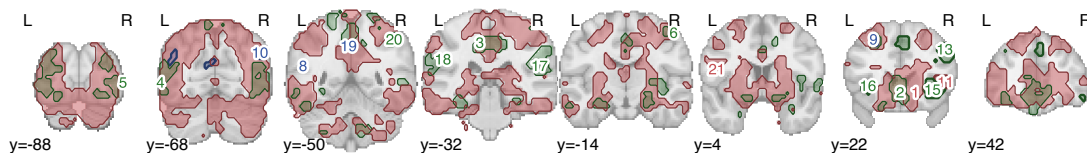
Surprise negative

no surviving clusters



All together

numbers = ROIs of page 1



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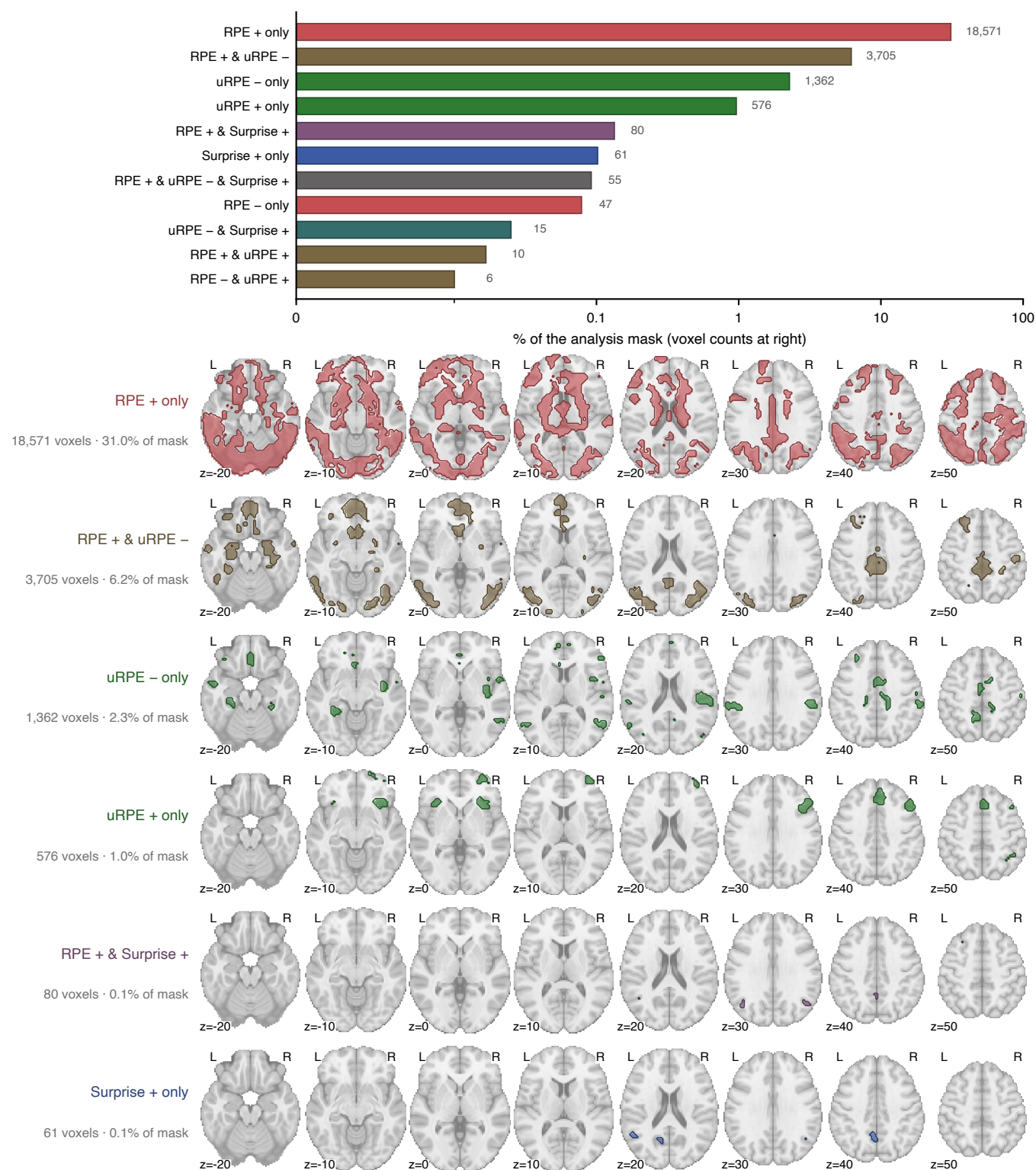
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d Which contrasts claim each piece of tissue

every combination, then the six biggest mapped



A voxel's category is the set of contrasts that call it significant, so "only" means no other contrast claims it. The remaining 5 combinations hold 133 voxels and are charted but not mapped.